

Yene Irvine

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TECHNICAL SKILLS

Frontend: TypeScript, JavaScript (ReactJS), TailwindCSS

Backend: Node.js (Express, NestJS), FastAPI, Java, REST & SOAP APIs

Cloud & DevOps: AWS, Azure, GCP, Vercel, Docker, Terraform, GitHub Actions, CI/CD pipelines

Data Engineering: SQL, ETL/ELT pipelines, Data Warehousing (Snowflake, DB2), Pandas, NumPy

Machine Learning: PyTorch, scikit-learn, matplotlib, WandB, HPC compute

EXPERIENCE

Data Engineer

Sep. 2024 – Present

Fuel Transport Inc.

Montreal, QC

- Built and integrated a Snowflake Rate Editor (FastAPI, React) to enable real-time editing of client rates, powering 100s of daily API calls and contributing to \$1.5M+ in revenue over 5 months.
- Developed and implemented a custom Azure Data Factory pipeline to migrate data from IBM DB2 to Snowflake, replacing Fivetran and saving \$40K+ annually in third-party costs.
- Optimized CI/CD pipelines (GitHub Actions, Docker, Azure Container Apps) to streamline deployment of the Snowflake Rate Editor's frontend and backend containers, improving reliability and reducing overhead.
- Leveraged Azure Monitor, Application Insights, and related Azure services to track API performance, analyze traffic patterns, and proactively resolve issues, ensuring high availability for enterprise clients.

Controls Engineer

Apr. 2019 – Jan. 2022

MCA Consultants

Charlottetown, PE & Halifax, NS

- Developed and commissioned 25 control narratives for utilities and HVAC control systems, integrating PLCs and process equipment, and working with contractors to enhance the Building Automation System.
- Created a program in Excel to determine relieving requirements for pressurized vessels based on NFPA and ASME standards; used in designing venting configurations for 50+ major vessels.
- Led a team to commission all process, utility, and HVAC systems and associated equipment under budget.

SELECTED PROJECTS

Detecting Breathing Irregularities with Deep Learning | 3D CNN, U-Net, PyTorch, HPC, WandB

- Developed and deployed a 3D Convolutional Neural Network with University of Calgary Biometrics Lab to detect sleep-related medical emergencies via thermal video, leveraging HPC compute and WandB for real-time tracking at the Calgary Drop-In Centre.

RunIt | React Native, NestJS, Firebase

- Built a mobile running app with real-time GPS tracking, Firebase authentication, and social features.

AI DJ Setlister | AWS EC2, Spotify Web API, OAuth2, Python

- Designed and deployed a web app hosted on AWS EC2 that uses Spotify's Web API and custom optimization algorithms to let users log in with their Spotify accounts and dynamically generate optimized DJ set playlists.

EDUCATION

University of Calgary

Calgary, AB

M.Eng., Software Engineering (GPA 3.94)

Graduated May 2024

Dalhousie University

Halifax, NS

B.Eng., Chemical Engineering (GPA 3.80)

Graduated Apr. 2018

CERTIFICATIONS

AWS Certified Cloud Practitioner CLF-C02 (2024)

Microsoft Azure Fundamentals AZ-900 (2024)